

Def. Let  $X$  be a Metric space,  $Y$  be a Normed space.

A sequence of functions  $\{f_n: X \rightarrow Y\}$  is called pointwise bounded on  $X$  if:

for any  $x \in X$ ,  $\{f_n(x)\}$  is bounded.

And it is called uniformly bounded if: for some  $M > 0$ ,

$$\sup_X |f_n| < M \text{ for all } n \in \mathbb{N}.$$

Thm. If  $\{f_n\}$  is a pointwise bounded sequence of complex-valued functions on a Countable set  $E$ , then  $\{f_n\}$  has a pointwise convergent subsequence.

Proof. Denote  $E = \{x_i \mid i \in \mathbb{N}\}$ . Since  $\{f_n(x_1)\}$  is bounded, it has a convergent subsequence, denote by  $\{f_{1,k}\}$ . Similarly,  $\{f_{1,k}(x_2)\}$  has a convergent subsequence, denote by  $\{f_{2,k}\}$ . Inductively, we obtain

$$\begin{array}{ccccccc} S_1: & f_{1,1} & f_{1,2} & f_{1,3} & \dots & & \\ S_2: & f_{2,1} & f_{2,2} & f_{2,3} & \dots & & \\ S_3: & f_{3,1} & f_{3,2} & f_{3,3} & \dots & & \\ & \vdots & & & & & \end{array} \quad \begin{array}{l} \searrow \text{sub} \\ \searrow \text{sub} \\ \searrow \text{sub} \\ \vdots \end{array} \quad \begin{array}{l} \text{Converges at } x_1 \\ \text{" } x_1, x_2 \\ \text{" } x_1, x_2, x_3 \\ \vdots \end{array}$$

And, the order of each function, is preserved regardless of they are contained any column, by the definition of subsequence. Then the subsequence

$$S = \{f_{n,n}\}$$

Converges for each  $x_i$ .  $\square$

Note:  $\mathcal{C}(X) = \{f: X \rightarrow \mathbb{C} \mid f \text{ continuous, bounded}\}$  where  $X$  is Metric space.

Lemma. If  $K$  is Compact Metric space,  $f_n \in \mathcal{C}(X)$ ,  $n \in \mathbb{N}$ , and  $f_n \rightarrow f$  on  $K$ , then  $\{f_n\}$  is equicontinuous on  $K$ .

Proof. Using  $\epsilon$ -method.  $f_n \rightarrow f \rightarrow$  Conti on Compact.

## Arzelà-Ascoli Theorem.

Let  $X$  be a Compact Metric space.

If  $\{f_n\}$  is pointwise bounded and equicontinuous on  $X$ ,

then  $\{f_n\}$  is uniformly bounded and  $\{f_n\}$  contains a uniformly convergent subsequence.

Proof. Given  $\varepsilon > 0$ , there exists  $\delta > 0$  s.t.  $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$  for all  $n \in \mathbb{N}$ .

Since  $X$  is compact, there are finitely many points  $p_1, \dots, p_r \in X$  s.t.  $X = \bigcup_{i=1}^r B_\delta(p_i)$ .

Since  $\{f_n\}$  is pointwise bounded, there exist  $M_i > 0$  s.t. for all  $n \in \mathbb{N}$ ,  $|f_n(p_i)| < M_i$ .

Now, for any  $x \in X$ , and  $n \in \mathbb{N}$ ,  $|f_n(x)| \leq |f_n(x) - f_n(p_i)| + |f_n(p_i)| \leq \varepsilon + \max\{M_i \mid i=1, \dots, r\}$ .  
for some  $p_i$  s.t.  $x \in B_\delta(p_i)$

To show that the second assertion, Compact + Metric  $\Rightarrow$  2nd-Countable  $\Rightarrow$  separable.

Let  $E \subseteq X$  be a Countable dense set. By the above Thm,

$\{f_n\}$  has a pointwise convergent subsequence on  $E$ , denote this by  $\{f_{n_i}\}$ . Simply,  $g_i = f_{n_i}$ .

Given  $\varepsilon > 0$ , take  $\delta > 0$  s.t.  $d(x, y) < \delta \Rightarrow |f_n(x) - f_n(y)| < \varepsilon$  for all  $n \in \mathbb{N}$ .

Since  $X$  is compact and  $E$  is dense in  $X$ , the set

$$\{B_\delta(x) \mid x \in E\} \text{ covers } X, \text{ and}$$

has a finite subcover,  $\{B_\delta(x_i) \mid i=1, \dots, m\}$ . Since  $\{g_i(x)\}$  converges for all  $x \in X$ ,

there exists  $N \in \mathbb{N}$  s.t.  $i, j \geq N, (1 \leq i \leq m) \Rightarrow |g_i(x_i) - g_j(x_i)| < \varepsilon$ .

And, for any  $x \in X$ , for some  $1 \leq i \leq m$ ,  $x \in B_\delta(x_i)$ , thus  $|g_i(x) - g_i(x_i)| < \varepsilon$  for all  $i \in \mathbb{N}$ .

Now, for all  $x \in X$  and  $i, j \geq N$ ,

$$|g_i(x) - g_j(x)| \leq |g_i(x) - g_i(x_i)| + |g_i(x_i) - g_j(x_i)| + |g_j(x_i) - g_j(x)| < 3\varepsilon.$$

